Fiscal Year:	FY 2013	**Task Last Updated:**	FY 04/14/2013
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Mapping Spaceflight-Induced Hypoxic Signaling And Response		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sgilroy@wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4009
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Botany		
PI Address 2:	430 Lincoln Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	06/12/2012	**End Date:**	06/11/2014
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	This research capitalizes on a rapid turn-around spaceflight opportunity and the Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFU) hardware to address how spaceflight affects gene expression related to hypoxic response in Arabidopsis. Root zone hypoxia is thought develop in spaceflight as weightlessness leads to a reduction in the buoyancy-driven convection that usually aids in gas exchange around organisms. This in turn leads to the development of oxygen-limiting conditions with adverse effects on plant vigor and yields. Our analysis of Ca2+ signaling responses and associated gene expression patterns in Arabidopsis indicates Ca2+ signaling is linked to anoxic response in plants. We have identified a suite of Ca2+-dependent proteins that show rapid transcriptional responses to anoxia in both roots and shoots. Mutants in two such genes, aca1-2, and cax2-2 (both Ca2+ transporters) are resistant to anoxic challenge. These mutants also show disrupted gravitropic response suggesting that Ca2+ signaling may provide a point of cross-talk between the anoxic and gravity response systems. We therefore propose to compare growth and gene expression patterns in aca1-2, cax2-2, and wild-type plants. Seedlings will be grown on orbit and fixed in RNAlater for subsequent analysis. Differences in growth will be quantified and correlated with transcription patterns analyzed post-flight by a combination of microarray and QPCR analyses. Comparison to ground-based controls maintained under normal and varying degrees of anoxic conditions at 1xG, or during clinorotation, will be used to ask how much of the responses seen on orbit can be explained by developing anoxia, disruption of gravisensing or a combination of anoxic challenge against the background of microgravity. Results from this analysis are expected to not only help advance our understanding of the development of anoxic response in space and on the Earth but also test whether the Ca2+ signaling system provides targets for genetically engineering potential countermeasures to this stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how low oxygen sensing operates in plants, targeting the Ca2+-dependent signaling machinery of the plant root. The results from this work will not only provide insight into how low oxygen levels impact on plant growth in space but also has impact on understanding similar responses on Earth. The low oxygen sensing/response systems triggered by anoxic conditions in space are also those employed by plants challenged by low oxygen on Earth, such as during flooding of the soil. This work aimed at analysis of how Ca2+ signaling is involved in the low oxygen response induced during spaceflight is also revealing a role for this system in this general flooding response. These results are therefore providing molecular targets for potential manipulation to tailor plant growth to varied soil oxygen status, such as promoting flooding tolerance in crop plants.
Task Progress:	This project has completed the development, flight, and sample recovery phases and sample analysis is currently underway. Preflight testing: The initial experimental setup to be used for this research was to develop a system that would operate within the 6 cm petri dish chambers used in the BRIC-PDFU flight hardware but that would permit use of dry seeds adhering to a filter wicking system. This would allow the transport of dry, highly resilient seeds to orbit and then injection of germination medium by the crew onboard the ISS to start the experiment. However, analysis of this system in the flight hardware revealed poor germination, likely due to an interaction between the wicking system setup and the closed sample environment of the Petri Dish Fixation Unit (PDFU) required for flight. An alternative experimental setup was therefore developed of using surface sterilized seeds plated on nutrient agar medium. Verification tests showed >90% germination of these seeds in the BRIC-PDFU hardware. Germination was reliably delayed by holding the plates at 4°C until the required time for seedling growth, which was started by warming the equipment to room temperature. Flight: This configuration was successfully launched using the SpaceX2 vehicle on March 1, 2013. The hardware consisted of two BRIC canisters, each containing five PDFUs and one HOBO temperature data logger. Plates of seeds (4 plates of wild-type and 3 each of two anoxia-related CAX2 Ca2+ transporter mutants) were integrated into the PDFUs on February 27, 2013. These were then placed in a double cold stowage bag maintained at 4°C and stowed into the SpaceX-2 Dragon capsule at Launch -24 hrs. This hardware was then successfully carried to the ISS where it was destowed from the Dragon capsule and transferred to the US Lab. After 8 days at room temperature, the crew injected RNAlater fixative into the PDFUs and the following day, the canisters were placed in the MELFI freezer onboard the ISS, stored at -95°C for 11 days, and then returned on March 24, 2013 in a -32°C Cold Bag using the Dragon capsule . These samples were deintegrated from the hardware on April 4, 2013 and transported frozen to the University of Wisconsin-Madison for further analysis Postflight analysis: Initial analysis revealed sufficient intact samples were recovered to allow for the gene expression analyses required from this experiment. Germination efficiency was >90% on all plates. Initial processing indicated the samples of seedlings would yield RNA in sufficient quantity and of high enough quality for both the quantitative PCR and whole genome gene expression analyses required for the project. Full characterization, extraction and analysis of these samples is now underway. Ground-based analysis: In parallel to the flight experiment, wild-type and the Ca2+ transporter mutants used in the spaceflight experiment have been monitored for their responses to gravitropic stimulation on Earth. Plants expressing the GFP-based 'Yellow Cameleon' Ca2+ sensor were generated and Ca2+ changes in response to gravitropic stimulation (90° rotation) monitored by confocal ratio imaging. Both mutants showed aberrant Ca2+ signals in the root tip to gravistimulation, with reduced onset of Ca2+ changes. These results suggest that the CAX2 Ca2+ transporter is involved in both the anoxia and gravity sensing systems of the plant root.
Bibliography Type:	Description: (Last Updated: 02/02/2026)

Articles in Peer-reviewed Journals	Toyota M, Gilroy S. "Gravitropism and mechanical signaling in plants." Am J Bot. 2013 Jan;100(1):111-25. http://dx.doi.org/10.3732/ajb.1200408 ; PubMed PMID: 23281392 , Jan-2013